Model-Informed Machine Learning for Macro-Finance Forecasting

Embedding RBC Structure into Neural Network Priors with Empirical Fine-Tuning

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i. Abstract

We embed a Real Business Cycle model into a neural network prior through pretraining on synthetic data, then fine-tune in two stages on cross-country macro data and US financial conditions. The network learns structural economic dynamics first and then adapts to real-world frictions. Each fine-tuning stage strictly improves forecast accuracy for the latent variables of interest, and the phase 2 network significantly outperforms AR(1) and BVAR benchmarks on the investment shock out-of-sample. Performance can only improve from here with richer models and more features, as well as better architectures. This paper provides evidence that structural economic theory and machine learning are complements with regards to forecasting capabilities.

ii. Introduction

Structural models are theoretically coherent but often misspecified. They impose strong assumptions about how economies work and struggle to fit the messy, frictional dynamics of real data. Machine learning models are flexible and data-driven but atheoretical and sample-hungry. Also, they are often fragile out of sample. Neither approach alone is adequate for a macro-finance forecasting problem.

This paper explores a combined approach. We embed a structural economic model directly into a neural network through pretraining on synthetic data, then adapt the network to real conditions through sequential empirical fine-tuning. The structural model, a two-shock Real Business Cycle (RBC) framework, provides the theoretical prior. The network learns this model prior to it ever seeing real data. Fine-tuning then updates the prior rather than replacing it, preserving structural coherence while accommodating the frictions and nonlinearities that the RBC model misses.

The approach addresses a fundamental problem in macroeconomics, in that structural models are derived from first principles but perform poorly as forecasting tools. A network with a structural prior should inherit the robustness of the theoretical model while gaining the empirical flexibility of a data-driven approach through fine-tuning. Applied, the estimated latent states are extremely relevant to asset pricing [8] and financial conditions. A model that can forecast these states out of sample has immediate applications in asset management and finance broadly [2].

The pipeline proceeds in three stages. First, we derive and solve a two-shock RBC model analytically, and generate a large set of synthetic simulations spanning a range of calibrated parameter values. A residual neural network is pretrained on these simulations, learning the structural transition dynamics of the model as its prior. Second, the network is fine-tuned on a cross-country panel of ten advanced economies at quarterly frequency from 1980Q1 to 2019Q4, exposing it to real macroeconomic variation across the original model's latent states, plus a set of macro features drawn from OECD databases, used to further inform the network. Third, the network is fine-tuned again on US-specific financial condition data, teaching the network the frictional, financially-driven dynamics of the actual US economy.

We evaluate the framework across five dimensions. Structural validity is assessed through impulse response functions comparing network rollouts against the analytical RBC solution. One-step forecast accuracy is benchmarked against AR(1), VAR(2), and BVAR. Multi-step rollout performance tests whether the structural prior prevents error accumulation at extended horizons. In-sample event studies covering the 2001 dotcom crash and the 2008 financial crisis assess whether the network has learned economically correct crisis dynamics. Finally, an out-of-sample evaluation covering 2023Q1 to 2025Q4 tests performance in a period the network has never seen, over the most aggressive monetary tightening cycle in four decades.

The results support the following conclusions. First, fine-tuning is necessary, as the pretrained network alone is uncompetitive on real data, and each fine-tuning stage steadily reduces MAE for the latent variables of interest. Second, the investment shock is the variable where the network consistently adds value, given the chosen augmentation features, with phase 2 outperforming all benchmarks on directional accuracy in-sample and maintaining positive R^2 out of sample while AR(1) and BVAR collapse. Third, the structural prior provides regime robustness that statistical models cannot replicate.

The remainder of the paper is organized as follows. Section iii. derives the RBC model and its analytical solution. Section iv. describes the calibration procedure. Section v. describes the synthetic data generation, as well as the pretraining and fine-tuning methodology. Section vi. evaluates the networks, and Section vii. concludes.

iii. Model

We employ a two-shock real business cycle model with both a total factor productivity shock and a capital/investment efficiency shock [9]. The model is solved analytically via log-linearization around the deterministic steady state. This yields a state-space representation whose transition dynamics serve as the structural prior for neural network pretraining.

The system is solved by separating stable and unstable eigenvalues and keeping only the stable path solution satisfying $\max|\lambda(P)| < 1$. The resulting state-space representation is:

$$s_{t+1} = Ps_t + Q\varepsilon_{t+1} \tag{1}$$

where P is the propagation matrix governing the persistence and spillovers of state variables, and Q is the impact matrix. The state vector is $s_t = \left[\hat{k}_t, \hat{c}_t, \hat{z}_t, \hat{\xi}_t\right]$, where the latent variables of interest are log deviations of capital stock, consumption, TFP shock, and capital/investment shock, respectively, from steady state values. The consumption function takes the linear form:

$$\hat{c}_t = a_k\hat{k}_t + a_z\hat{z}_t + a_\xi\hat{\xi}_t \tag{2}$$

where the coefficients are solved analytically as functions of the structural parameters. See Table 1 for definitions of all variables included in the model.

iv. Calibration

The model is calibrated via Simulated Method of Moments (SMM) by matching a set of empirical business cycle moments computed from US quarterly data. Three parameters are fixed at standard values from the macro literature before estimation. The remaining five parameters $\theta = (\phi, \rho_z, \rho_\xi, \sigma_z, \sigma_\xi)$, denoting disutility of labour, shock persistence, and shock volatility respectively, are estimated by minimizing the weighted distance between model-simulated moments and their empirical counterparts.

Fixed Parameters.— The capital share $\alpha = 0.33$, discount rate $\beta = 0.99$, and depreciation rate $\delta = 0.025$ are fixed at values standard in the quarterly RBC literature [10].

Simulated Method of Moments.— Empirical moments are constructed from FRED data [6] for US real GDP (GDPC1), real personal consumption expenditures (PCECC96), and gross private domestic investment (GPDIC1) over the period 1960Q1–2019Q4. The target moment vector has eight moments: the steady-state consumption-to-output ratio, the standard deviations of output, consumption, and investment growth, the relative volatility for output, consumption, and investment,

the first-order autocorrelation of the HP-filtered output cycle [7], and the first-order autocorrelation of investment growth. The weighting matrix W places a higher weight on the shock persistence and investment volatility moments, which are the dimensions most directly informative about shock persistence and stochasticity. The SMM estimator solves:

$$\hat{\theta} = \arg\min_{\theta} \left(m^{model}(\theta) - m^{data}\right)^T W \left(m^{model}(\theta) - m^{data}\right) \tag{3}$$

where the model moments are computed by simulating the model at parameters θ for a large number of periods and computing the corresponding moments from the simulated series. Table 2 reports the full parameter vector. Table 3 reports the moment fit and compares empirical and model-simulated moments at the estimated parameters.

v. Methodology

The pipeline has three stages:

Pretraining.— We generate 50,000 synthetic simulations of 250 quarters each from the RBC model, drawing parameter vectors uniformly from a range centred on the SMM estimates. A residual MLP with input dimension 19, hidden dimension 512, four residual blocks, and 2.1 million parameters is pretrained on 12.5 million consecutive-quarter pairs. Each input vector includes 15 auxiliary slots for empirical features added during fine-tuning. In pretraining, these slots carry uniform random noise drawn from the empirical ranges of candidate real-world features. Example features are credit spreads, policy rates, inflation, unemployment, and habit-formation consumption residuals. Using noise rather than zeros supplies non-degenerate variation without any predictive content, so gradients drive the network to treat these channels as uninformative while keeping the input projections well-conditioned.

In place of plain MSE, we impose a two-term loss [1] that penalizes violations of the RBC model's structural equations, as well as penalizing deviations of the implied quantities from the simulated realized values. This enforces structural consistency alongside prediction accuracy. Denoting the network's outputs $\widehat{k_{pred}}, \widehat{z_{pred}}, \widehat{\xi_{pred}}$, the next-period consumption, output, and investment are reconstructed from the policy function and model identities as follows:

$$\widehat{c_{pred}} = a_k \widehat{k_{pred}} + a_z \widehat{z_{pred}} + a_\xi \widehat{\xi_{pred}} \tag{4}$$

$$\widehat{y_{pred}} = \frac{(\phi+1)\widehat{z_{pred}} + \alpha(1+\phi)\widehat{k_{pred}} - (1-\alpha)\widehat{c_{pred}}}{\phi+\alpha} \tag{5}$$

$$\widehat{i_{pred}} = \left(\frac{1}{1-\frac{c}{y}}\right)\widehat{y_{pred}} - \left(\frac{\frac{c}{y}}{1-\frac{c}{y}}\right)\widehat{c_{pred}} \tag{6}$$

where a_k, a_z, and a_ξ are the analytical policy coefficients recovered from the solver for each simulation's parameter vector. Then, the loss terms are computed as:

$$L_{\text{controls}} = E\left(|\widehat{c_{\text{pred}}} - \widehat{c_{t+1}}| + \left|\widehat{i_{\text{pred}}} - \widehat{i_{t+1}}\right|\right) \tag{7}$$

$$L_{\text{dynamics}} = E\left(\left|\widehat{k_{\text{pred}}} - \left((1-\delta)\widehat{k_t} + \delta\widehat{i_t} + \delta\widehat{\xi_t}\right)\right|\right) \tag{8}$$

$$\text{Total Loss} = L_{\text{controls}} + L_{\text{dynamics}} \tag{9}$$

We implement a 90/10 train/validation split under a fixed seed. We use AdamW with initial learning rate 10^{-3}, weight decay 10^{-4}, batch size 2048, cosine annealing over 50 epochs, and a single GPU.

Stage 1 Fine-Tuning.— The pretrained network is fine-tuned on a cross-country panel of ten advanced economies (Canada, Denmark, Finland, Japan, Netherlands, Norway, Sweden, Switzerland, UK, USA) at quarterly frequency from 1980Q1 to 2019Q4, yielding 1,590 training pairs. State variables $(\hat{k}, \hat{c}, \hat{z}, \hat{\xi})$ are constructed from Penn World Tables v11.0 [5], interpolated to quarterly via cubic spline and HP-filtered ($\lambda = 1600$) [7]. The investment shock is recovered as the consumption function residual. Ten macroeconomic features from OECD databases [13] are appended as conditioning information: GDP growth, unit labour cost growth, CPI inflation, unemployment, and the yield spread, each entered as a level and an eight-quarter rolling standard deviation, representing volatility. Countries are presented sequentially in order of data coverage; each country inherits the weights from all prior countries, implementing a regularized sequential transfer across the panel.

Stage 2 Fine-Tuning.— Five US-specific financial features replace the noise-padded input slots: the BAA–AAA corporate spread, Federal Funds Rate, $log(VIX)$, the 10yr–FFR term spread, and the SP 500 quarterly log return, all sourced from FRED [6] over 1990Q1–2019Q4 (120 observations). These span the main dimensions of US financial conditions, being credit risk, rate level, uncertainty, curve shape, and equity market state. The network is fine-tuned on US data only, learning how financial frictions interact with the RBC state dynamics in a single institutional context. Learning rate is 10^{-5}, an order of magnitude below Stage 1, ensuring only modest updates to already-adapted weights.

vi. Results and Evaluation

We evaluate the network across five dimensions being structural validity, one-step forecast accuracy relative to time-series benchmarks, multi-step rollout performance, in-sample crisis event studies, and an out-of-sample test window. Benchmarks are AR(1), VAR(2), and BVAR with Minnesota prior [3, 11], each fitted on the same 1990Q1–2013Q4 training window. All models target the same three state variables: capital stock, TFP shock, and investment shock.

Structural Validity.— We verify that fine-tuning has not destroyed the structural prior by computing impulse response functions (IRFs) following a one standard deviation TFP shock from steady state and comparing network rollouts against the analytical solution of the linearized RBC model. See Figure 1. The pretrained network produces IRFs broadly consistent with the analytical solution for capital stock and TFP shock. Fine-tuning introduces larger deviations from the analytical benchmark. This reflects the partial updating of the structural prior by real data dynamics that are not well-described by the frictionless RBC framework. This partial erosion of the structural prior is an expected consequence of fitting to real data. The relevant question is whether it retains sufficient structural coherence, which we ascertain it does given that all variables mean revert following the shock.

One-Step Forecast Accuracy.— Table 4 reports MAE, RMSE, and R^2 for all models on the held-out validation set from 2014Q1 to 2019Q4. See Figure 2.

VAR(2) achieves the lowest MAE for capital stock and TFP shock, and the highest $\mathbf{R^2}$ across all variables. This is not surprising as the 2014 to 2019 validation window is a period of smooth expansion, which allows high autocorrelation models to excel. The pretrained network performs extremely poorly on all variables. This confirms that the structural prior alone is insufficient for real-data forecasting without empirical calibration. Fine-tuning substantially corrects this, as phase 1, and phase 2 to an extent, significantly reduce MAE across all variables.

This demonstrates that both broad macroeconomic data and US financial condition features add forecasting capabilities for the network.

Multi-Step Rollout and Regime Robustness.— Figure 3 reports multi-step MAE at horizons of one, four, and eight quarters for the neural network stages.

The pretrained network degrades sharply with horizon for capital stock and TFP shock, with eight-quarter MAEs approximately twenty times those of the fine-tuned models. Phase 1 and phase 2 exhibit broadly flat error profiles for these variables. This indicates that the structural prior prevents runaway error accumulation even at extended horizons.

In-Sample Event Studies.— To assess crisis dynamics, we roll each model forward from the onset of the 2001 dotcom crash (2000Q1) and the 2008 financial crisis (2007Q1), initializing from observed states and using actual financial features throughout. Both events fall within the training window and these simulations are therefore structural validation exercises. The relevant question is not whether the network tracks the crisis, since it has seen the data, but whether the phase 2 improvement over phase 1 is concentrated in the quarters where the financial condition features carry the most signal. We observe that the phase 2 advantages are largest when credit conditions deteriorate most sharply, so we infer that the BAA-AAA spread and VIX are doing economically meaningful work rather than statistical overfitting. See Figure 4.

The pretrained network diverges immediately in both episodes. Phase 1 tracks capital stock and TFP shock trajectories but produces flat predictions for the investment shock. For the 2008 crisis, phase 2 captures the sharp decline in the investment shock around quarters eight through ten and recovers both the shape and approximate magnitude of the actual trajectory while phase 1 remains flat throughout. This improvement is concentrated in exactly the quarters when the BAA-AAA spread and VIX spiked most dramatically. If phase 2 were simply overfitting to 2008 rather than learning that credit spreads predict investment collapses, we would not expect its advantage over phase 1 to be so precisely aligned with the timing of financial distress. The 2001 episode is somewhat less successful for all models, consistent with that recession being driven by an equity correction rather than a broad credit tightening. However, better performance in the 2001 domain could be achieved with a different feature augmentation strategy.

Out-of-Sample Evaluation.— We evaluate the phase 2 network on eleven out-of-sample quarters, 2023Q1 to 2025Q4. State variables are reconstructed from FRED using the same pipeline as training. The HP filter is applied to the full history back to 1980Q1 to mitigate end-of-sample bias. See Table 5 and Figure 5. VAR(2) is not included in this analysis as it exhibits immense standard errors, given that it is essentially conditioning on 2 quarters of data, and then just rolling forward it's own predictions, which gives extremely divergent behaviour.

AR(1) and BVAR outperform all networks on capital stock, consistent with capital being too smooth to benefit from richer conditioning information. The network is effectively conditioning on TFP, investment shock, and other financial features to predict where capital goes, and this extra conditioning information just adds noise rather than signal; in short, the network is overthinking it. For the TFP shock, all three network versions beat AR(1) and BVAR, with the pretrained model achieving the lowest MAE. This suggests the RBC structural prior captures TFP dynamics well across cycles. For the investment shock, the fine-tuning progression is preserved out of sample, as phase 2 achieves the lowest MAE and best RMSE, while the BVAR and AR(1) produce a deep negative R^2. The AR(1)'s poor OOS performance for the investment shock reflects a large initialization error at the start of the window when the OOS starting state differs substantially from the training-sample mean. This is a regime-boundary problem that the network handles much better by conditioning on contemporaneous financial features.

These results should be interpreted cautiously given the short OOS window of eleven quarters. Nevertheless the consistent ordering of phase 2 performing better than phase 1, performing better

than the pretrained network for the investment shock across both in-sample and out-of-sample windows provides meaningful evidence that the US financial condition features contribute significant forecasting capability, especially when compared with classical statistical models.

Summary.— Four findings emerge. First, fine-tuning is necessary: the pretrained network alone is far from competitive with any benchmark on real data, with the exception of the TFP shock out of sample. Second, VAR(2) and other statistical models dominate one-step forecasting in calm periods for $\hat{k}$ and $\hat{z}$, consistent with the high persistence of these variables during the 2014–2019 expansion, but completely fall apart when noisier dynamics exist. Third, the investment shock $\hat{\xi}$ is the variable for which the network most consistently adds value: phase 2 outperforms AR(1) and BVAR on directional accuracy and R^2 in-sample, and maintains positive explanatory power out of sample where the AR(1) fails. Fourth, the structural prior provides regime robustness that statistical benchmarks lack. It also prevents error accumulation in multi-step rollouts, stabilizes forecasts at distributional boundaries, and captures the 2008 investment shock dynamics that VAR-style models cannot reproduce without re-estimation.

vii. Conclusion

This paper proposes and implements a model-informed machine learning framework for macro-finance forecasting. Starting from a two-shock Real Business Cycle model solved analytically, we pretrain a residual neural network on a large set of synthetic simulations to embed the structural dynamics of the model as a prior over the network's weights. The network is then fine-tuned in two stages: first on a cross-country panel of ten advanced economies from 1980Q1 to 2019Q4, and then on US-specific financial condition data spanning 1990Q1 to 2019Q4. Each stage is designed to bring the network closer to real empirical dynamics while preserving the structural coherence instilled during pretraining.

The central claim of this paper is that the proposed three-stage pipeline of pretraining a network on theory, fine-tuning on macro, then fine-tuning on finance produces a network that is both structurally informed and empirically adaptive. The evidence supports this claim. The pretraining stage successfully instils a structural prior and the network learns to predict next-period states somewhat consistent with RBC dynamics. The first fine-tuning stage then adapts this prior to real macroeconomic variation across ten countries. The second fine-tuning stage refines the network further for the US, with the BAA-AAA credit spread, federal funds rate, VIX, term spread, and equity returns providing financial condition signals that meaningfully improve forecasting both in-sample and out-of-sample.

The improvement is monotone across stages, in that each fine-tuning stage strictly reduces mean absolute error relative to the previous. This monotone progression holds on the held-out validation set, on the in-sample event studies, and on the out-of-sample window. The consistency of this ordering across evaluation environments suggests that the structural prior is not being overwritten by fine-tuning, and that it is being refined and complemented by empirical information.

The results reported here should be understood as a proof of concept and a lower bound on what the framework can achieve. The current implementation makes a number of deliberately conservative choices. The network architecture is a standard residual MLP with no attention mechanism and no recurrent structure. The pretraining data is generated from a two-shock linearized RBC model. The fine-tuning feature set is somewhat limited.

Each of these features represents a dimension along which the framework can be extended. A richer structural model at the pretraining stage such as a medium-scale DSGE with nominal rigidities [12, 4] would provide a more informative prior. A larger conditioning feature set would

give the network richer information at each quarter. A transformer architecture trained on multi-quarter windows would allow conditioning on the full history of states rather than only the current period.

The broader implication is that structural economic theory and modern machine learning are complements rather than substitutes in the forecasting problem. The pretraining to fine-tuning approach addresses the standard critiques of ML in economics, since the structural prior provides theoretical grounding, and the synthetic pretraining data alleviates the small-sample problem. Conversely, the results also demonstrate the limits of structural models used alone, since the pretrained network, despite encoding correct theoretical dynamics, is far from competitive with even an AR(1) on real data without empirical fine-tuning.

We view this framework as a foundation for broader research. Deriving the theoretical prior, embedding it via pretraining, and then refining it empirically can accommodate richer models and larger datasets, as well as more sophisticated architectures without modification to the overall design. The present paper establishes that the structural prior survives empirical fine-tuning, and that the resulting network adds significant forecasting value.

All code used in this paper is available at `https://github.com/pepogara/tl_framework`.

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Appendix

Table 1: Notation for All Model Variables.

Variable	Definition
$\hat{c}_t$	Log-deviation of consumption from steady state
$\hat{k}_t$	Log-deviation of capital stock from steady state
$\hat{n}_t$	Log-deviation of labour supply from steady state
$\hat{z}_t$	TFP shock (log-deviation)
$\hat{\xi}_t$	Capital/investment efficiency shock (log-deviation)
$\hat{i}_t$	Log-deviation of investment from steady state
$\hat{y}_t$	Log-deviation of output from steady state
$\hat{r}_t$	Log-deviation of return to capital from steady state

Table 2: Model Parameters Estimated via SMM.

Parameter	Definition	Value
α	Capital share in production	0.3300
β	Household discount rate	0.9900
δ	Capital depreciation rate	0.0250
φ	Frisch elasticity of labour supply (inverse)	1.2016
ρ_z	TFP shock persistence	0.7285
ρ_ξ	Investment shock persistence	0.8558
σ_z	TFP shock volatility	0.0057
σ_ξ	Investment shock volatility	0.1000
c/y	Steady-state consumption-to-output ratio	0.6435

The Consumption-to-Output Ratio is the Empirical Sample Mean Used to Calibrate the Steady State.

Table 3: SMM Moment Fit.

Moment	Data	Model	Difference
$E[c/y]$	0.6435	0.6435	0.0000
$\text{std}(g^y)$	0.0105	0.0094	-0.0011
$\text{std}(g^c)$	0.0103	0.0089	-0.0013
$\text{std}(g^i)$	0.0405	0.0363	-0.0042
$\text{std}(g^c)/\text{std}(g^y)$	0.9760	0.9484	-0.0276
$\text{std}(g^i)/\text{std}(g^y)$	3.8455	3.8522	$+0.0067$
acf(y-cycle)	0.7721	0.8354	$+0.0633$
$\text{acf}(g^i)$	0.1335	-0.0792	-0.2126

Figure 1: Impulse Response Functions Following a One-Standard-Deviation TFP Shock.

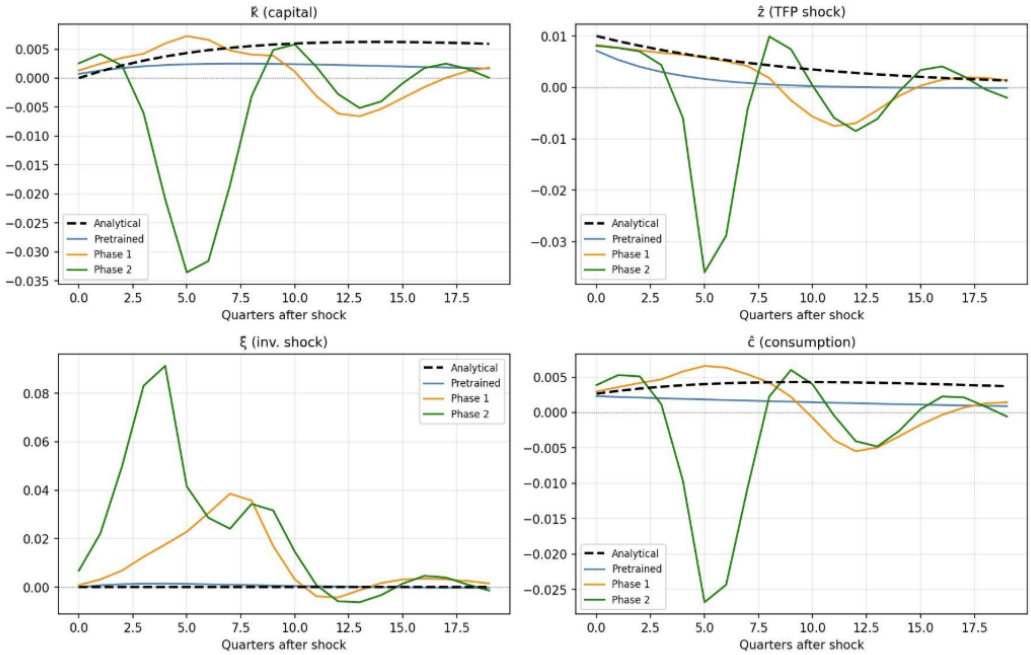

Table 4: One-Step-Ahead Forecast Metrics 2014Q1 to 2019Q4.

Variable	**Model**	**MAE**	**RMSE**	$\mathbf{R^2}$
$\hat{k}$ (capital)	AR(1)	0.00030	0.00039	+0.706
	VAR(2)	0.00015	0.00021	+0.930
	BVAR	0.00032	0.00038	+0.638
	Pretrained	0.04307	0.04841	−4980
	Phase 1	0.00412	0.00513	−55
	Phase 2	**0.00054**	**0.00061**	**−0.166**
$\hat{z}$ (TFP)	AR(1)	0.00043	0.00054	+0.807
	VAR(2)	0.00021	0.00028	+0.964
	BVAR	0.00057	0.00068	+0.731
	Pretrained	0.12515	0.12754	−10506
	Phase 1	0.00443	0.00545	−18
	Phase 2	**0.00089**	**0.00101**	**+0.337**
ξ (inv. shock)	AR(1)	0.03217	0.05418	+0.503
	VAR(2)	0.01888	0.02341	+0.847
	BVAR	0.03303	0.04102	+0.518
	Pretrained	0.09819	0.10991	−1.06
	Phase 1	0.09251	0.10485	−0.87
	Phase 2	**0.04281**	**0.06148**	**+0.357**

Figure 2: One-Step-Ahead Forecast Accuracy.

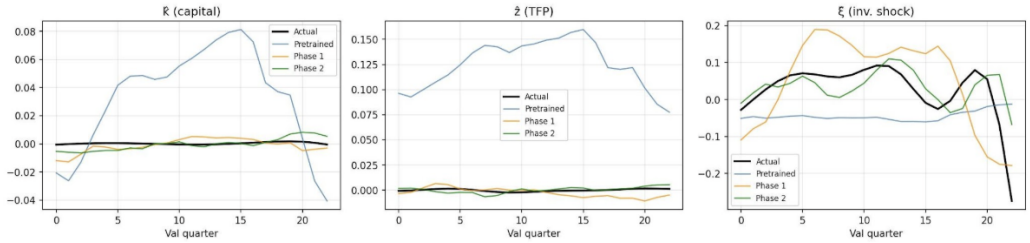

Figure 3: Multi-Step Forecast MAE at 1, 4, and 8 Quarter Horizons.

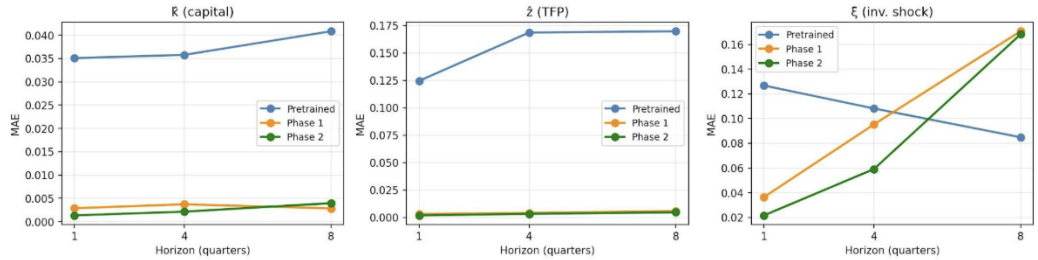

Figure 4: Event Study Rollouts for the 2001 DotCom Crash and the 2008 Financial Crisis.

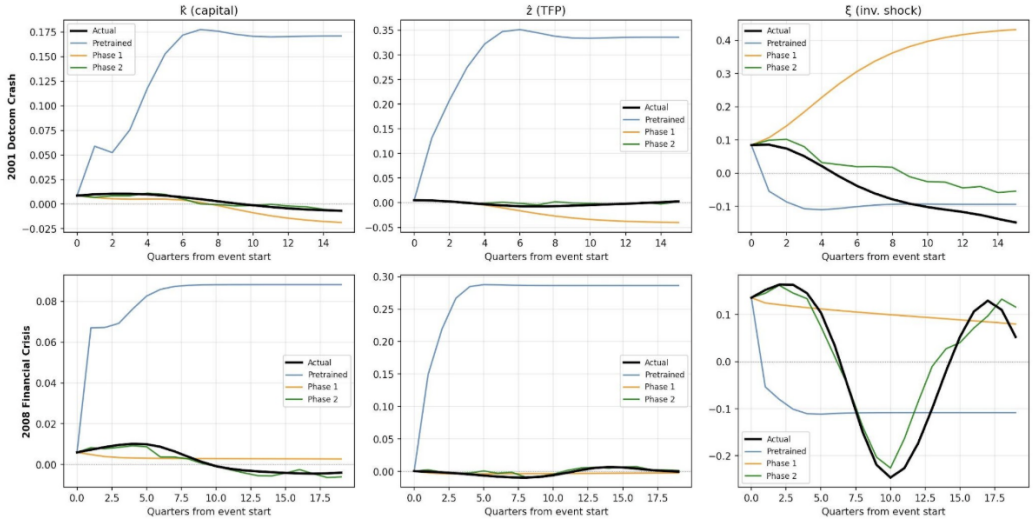

Table 5: Out-of-Sample Forecast Metrics, 2023Q1 to 2025Q4.

Variable	Model	MAE	RMSE	R^2
$\hat{k}$	AR(1)	0.00107	0.00119	+0.344
	BVAR	0.00150	0.00176	−0.43
	Pretrained	0.00219	0.00238	−1.61
	Phase 1	0.00274	0.00296	−3.04
	Phase 2	**0.00251**	**0.00272**	**−2.42**
$\hat{z}$	AR(1)	0.00404	0.00468	−0.77
	BVAR	0.00525	0.00634	−2.26
	Pretrained	0.00233	0.00301	+0.265
	Phase 1	0.00277	0.00338	+0.073
	Phase 2	**0.00276**	**0.00348**	**+0.020**
ξ	AR(1)	0.05339	0.09800	−4.84
	BVAR	0.07638	0.12094	−7.89
	Pretrained	0.03476	0.03962	+0.046
	Phase 1	0.03237	0.03995	+0.030
	Phase 2	**0.02890**	**0.03687**	**+0.174**

Figure 5: Out-of-Sample Forecasts Versus Actuals, 2023Q1 to 2025Q4.

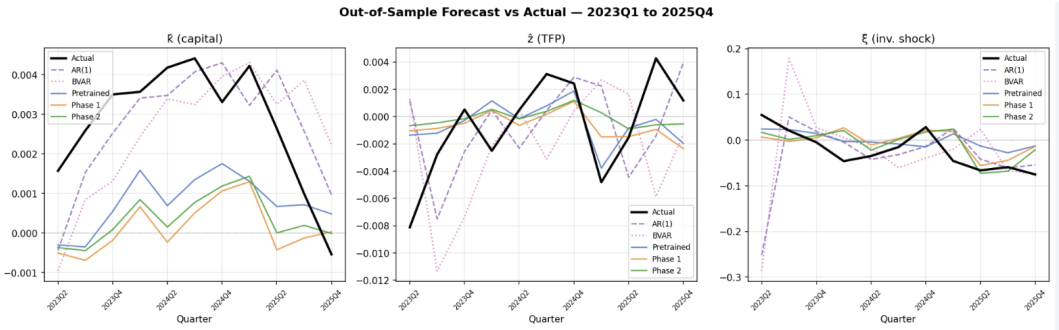

Table 6: Sampling Range for the Synthetic Generated Data.

Parameter	Calibrated Value	Sampling Range
α	0.3300	$[0.33 \cdot 0.9,\ 0.33 \cdot 1.1]$
ρ_z	0.7285	$[0.7285 \cdot 0.95,\ \min(0.999,\ 0.7285 \cdot 1.05)]$
ρ_ξ	0.8558	$[0.85,\ 0.95]$
φ	1.2016	$[1,\ 1.5]$
c/y	0.6435	$[0.6435 \cdot 0.95,\ \min(0.95,\ 0.6435 \cdot 1.05)]$
β	0.9900	$[0.985,\ 0.995]$
δ	0.0250	$[0.02,\ 0.03]$
σ_z	0.0057	$[0.01,\ 0.03]$
σ_ξ	0.1000	$[0.01,\ 0.03]$